

HENOK WEHIBE

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WORK EXPERIENCE

FastAPI-Based ANPR System

Systems Architect & Developer

2024–Present

Architected and delivered a secure, self-hosted Automatic Number Plate Recognition (ANPR) platform with hybrid edge/cloud processing, production deployment automation, and real-time analytics for parking, access control, and fleet monitoring.

Built the application and infrastructure layer for video ingestion, plate recognition, permit validation, job tracking, and reporting.

- Designed FastAPI services for real-time ANPR processing, event-driven workflows, authenticated APIs, and operational dashboards.
- Engineered hybrid OCR pipelines combining edge/mobile inference with cloud verification to improve latency and accuracy for live deployments.
- Curated model/data workflows for dataset preparation, training validation, and performance tuning across UK registration plate formats.
- Automated secure deployments with Docker, Ansible, Linux hardening, and operational monitoring for production environments.
- Developed comprehensive Ansible infrastructure automation for AWS EC2/Lightsail provisioning, server configuration, and application deployment with 15+ roles.
- Built Flutter mobile applications (ANPR Edge OCR & Hybrid OCR) for on-device vehicle detection and license plate recognition using Google ML Kit and TFLite.
- Implemented offline SQLite queueing with automatic sync, on-device plate cropping, and real-time streaming to FastAPI backend.
- Created security testing harness with pytest for defensive security validation including TLS enforcement, authentication, and upload validation.
- Deployed production monitoring with Monit, Fail2ban, logrotate, and PAM-based login alerts for operational visibility.

MicrometreUK

Lead / Full-Stack Developer

2022–2024

Led the design and delivery of MicroANPR.com, a cloud-native, multi-tenant ANPR SaaS platform for parking operators, gated communities, fleet management, and retail car parks. Built the product from concept through production, spanning backend services, frontend dashboards, ML pipelines, and DevOps automation.

Built the core platform for concurrent camera streams, multi-tenant operations, secure API integrations, and live reporting.

- Implemented high-throughput Go services with Gin and PostgreSQL to process thousands of concurrent camera streams with low memory overhead.
- Designed multi-tenant access control, webhook integrations, and dashboard APIs for real-time parking operations and fleet workflows.
- Developed the computer vision pipeline with OpenCV, PaddleDetection, and PaddleOCR for UK plate recognition and data preprocessing.
- Deployed and automated production infrastructure using Docker, Ansible, Nginx, Cloudflare, and Linux security hardening.

Projects

1. **Browser-Based AI Chat (WebLLM)** - Built a 100% client-side AI chat application using WebLLM and WebGPU with zero server dependencies.
 - Implemented WebGPU for hardware-accelerated AI inference directly in the browser.
 - Demonstrated modern JavaScript and AI/ML integration techniques.
1. **Daily Cashier** - Privacy-first sales management system using WebAssembly and SQLite.
 - Built offline-first application with OPFS (Origin Private File System) for persistent client-side storage.
 - Developed with Next.js and SQLite WASM, requiring zero server dependencies.
1. **RAG Quiz System** - Retrieval-Augmented Generation implementation using Ollama.
 - Integrated embeddings and language models for intelligent question-answering.
 - Built with Python for backend processing and vector storage.
1. **Gemma Web & Semantic Quiz** - In-browser implementation of Google Gemma AI and quiz systems using WebGPU and Transformers.js.
 - Built client-side intelligent quiz systems that grade answers using semantic similarity.
 - Showcased advanced ML deployment by running models entirely in the browser, reducing server costs to zero.
1. **AI-Powered Smart Cameras** - IoT computer vision solutions (Smart Camera, Plants Camera).
 - Developed edge pipelines for real-time object detection and tracking plant health, growth, and environmental factors.
 - Engineered solutions for continuous video processing and ultra-low framerate capture.
1. **DevOps & Infrastructure Projects**
 - Developed Ansible roles for automated WireGuard VPN deployment and event-driven infrastructure.
 - Configured Kubernetes clusters (kubeadm, Microk8s) with Calico/Flannel networking and Nginx Ingress.
 - Built custom Debian/Linux distributions using Packer for consistent development environments.
 - Set up private Docker Registry and Gitea instances on AWS, Azure, and GCP.
 - Implemented Nginx/HAProxy reverse proxy with SSL termination and Fail2ban security.
 - Created monitoring systems with Monit, Crontab, and Git for configuration tracking.

Freelance/Self-Employed
Full-Stack Web Developer

2020-2022

Developed e-commerce websites, landing pages, and blogs for clients across various industries.

- Built custom WordPress themes and React/Next.js applications from initial concept to deployment.
- Delivered end-to-end solutions: domain registration, email hosting, SSL certificates, and ongoing maintenance.
- Achieved significant performance improvements using Edge Computing (Cloudflare, Vercel) and code optimization.
- Conducted comprehensive audits for performance, accessibility, and SEO.

Built custom web applications and business tools for small businesses.

- Implemented store locators, interactive maps, and custom features using Google Maps API.
- Configured hosting infrastructure with Nginx, SSL termination, and automated deployments.

TECHNICAL SKILLS

Languages	Go, Python, JavaScript/TypeScript
Frontend	React, Next.js, Astro, Tailwind CSS, Framer Motion
Backend	Gin (Go), FastAPI, Node.js, Express, REST APIs, SSE
Computer Vision	PaddleDetection, PaddleOCR, PyTorch, TensorFlow, OpenCV
AI/ML	WebLLM, RAG, Ollama, Embeddings, llama.cpp
Databases	PostgreSQL, MySQL, Redis, MongoDB, SQLite (WASM)
Cloud Platforms	AWS, Google Cloud Platform, Microsoft Azure, Cloudflare
DevOps	Docker, Ansible, Kubernetes, Nginx, HAProxy
Tools	Git, Packer, Vagrant, FFmpeg

CERTIFICATIONS

FreeCodeCamp	Responsive Web Design
FreeCodeCamp	JavaScript Algorithms and Data Structures

EDUCATION

<i>South Thames College</i>		<i>London, UK</i>
Diploma	Information Technology	
GCSE	Triple Science (Biology, Chemistry, Physics), English, Maths	